

INTEGRATED ELECTRICAL SYSTEMS OF SOLAR PHOTOVOLTAIC TECHNOLOGY

PROJECT

GROUP 2



TÉCNICO LISBOA

Installation of a PV system in Olhão's City Hall

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2025/2026 – 2nd Semester, P3

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1 Introduction

This report presents the design of a solar photovoltaic system with energy storage for Olhão's City Hall. Nowadays, making use of solar energy is a practical way to reduce electricity costs and promote sustainability. To create a system that works well and is economically viable, it is necessary to choose the right equipment.

Therefore, this report will analyse and compare different commercial models of solar panels, inverters, and batteries. The comparison will focus on finding the best balance between performance and cost.

The main objective of this project is to maximise energy production during the day in order to reduce grid import costs, and to size a battery system capable of sustaining the building's base loads during the night.

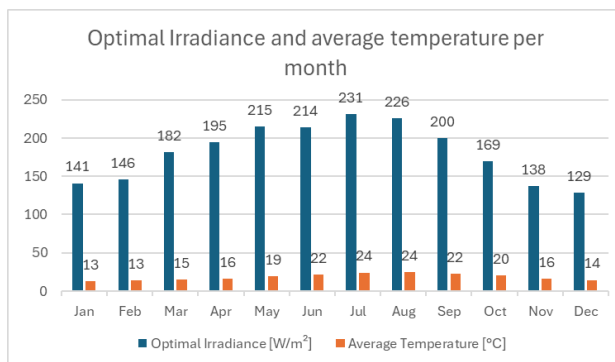
2 Site characterization

The Olhão City Hall is located at the coordinates 37°01'30.1"N 7°50'30.6"W, Faro, Portugal. It is a 2-floor building, and it features a flat roof with an available area of 402.50 m^2 , highlighted in green, Figure 1. The site has no surrounding obstacles that could cause shading in the PV system, ensuring that the maximum solar irradiation is captured throughout the day.

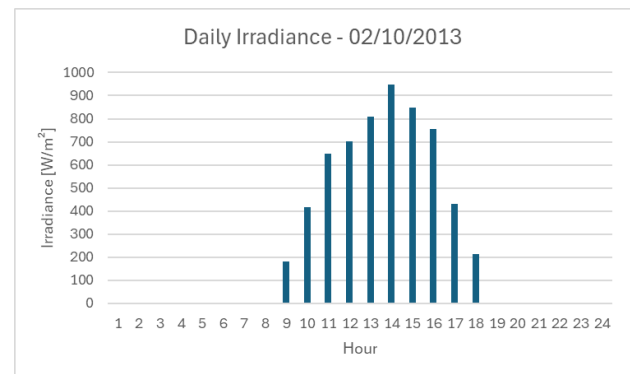


Figure 1: Site top view and available area

Beyond the physical topology of the roof, the viability of a photovoltaic project also relies on the local climatic conditions, such as Irradiance (G) and Temperature (T). The data from Figure 2a was obtained from *PVGIS*, and it represents the local average irradiance and temperature values in a 19-year interval (2005-2023). These values were extracted assuming a surface slope angle of 33°, corresponding to the optimum slope angle automatically calculated by the software to maximise energy yield.



(a) 19-year average irradiance and temperature.



(b) Daily irradiance profile (02/10/2013).

Figure 2: Comparison between the 19-year average and a typical winter daily profile in Olhão

Understanding these two parameters is essential for accurately sizing the PV system. Irradiance (G) directly determines the generated current, meaning that higher solar radiation leads to a higher power output. The ambient temperature has an inverse effect on performance. As the solar panels heat up, their open-circuit voltage (V_{oc}) decreases, leading to a decrease in the overall efficiency due to thermal losses. This location has a particular advantage: The Algarve region receives the highest levels of solar irradiance in Portugal.

Figure 2a shows the 19-year average irradiance. Since these values are averaged over the full sunlight periods, the overall irradiance values appear relatively low (around $200\text{W}/\text{m}^2$). However, to demonstrate the actual power available during sunlight hours, Figure 2b demonstrates a daily irradiance profile for a day in February. Even in winter, the available sunlight reaches a peak of $947\text{W}/\text{m}^2$, so this confirms that the system will maintain a great energy generation capacity throughout the entire year.

3 Load Curve

To accurately size the PV system, a detailed estimation of the building's energy consumption was performed. Based on the facility's profile, the Olhão City Hall has an estimated annual energy demand of approximately 64.9 MWh. The monthly distribution of this demand, as well as the specific load categories, is demonstrated in Figure 3.

Camara Municipal de Olhao -- Estimated Monthly Load Profile (800 m2 occupied)																
Load category	Appliance	Type	Installed kW	January	February	March	April	May	June	July	August	September	October	November	December	Annual kWh
Lighting	Lighting	workday	6,40	1166,5	989,2	973,3	844,8	842,2	752,6	811,1	851,4	893,6	949,2	1079,8	1134,9	11289
Office equipment	Office equipment	workday	6,00	910,1	849,0	928,2	853,1	903,6	861,3	928,2	913,4	902,3	905,2	882,6	885,5	10723
Break room	Break room / kitchenette	workday	13,75	802,2	748,3	818,1	751,9	756,6	683,2	695,4	684,3	715,8	797,9	778,0	780,5	9012
Refrigeration	Office fridge 1	24h	0,15	68,3	62,2	78,1	80,3	87,9	94,5	97,6	97,6	89,8	83,0	70,9	68,3	979
	Office fridge 2	24h	0,15	68,3	62,2	78,1	80,3	87,9	94,5	97,6	97,6	89,8	83,0	70,9	68,3	979
Heating (HVAC)	Heating (HVAC)	seasonal	8,00	1586,9	1332,2	971,0	297,5	0,0	0,0	0,0	0,0	157,3	473,5	1231,2	1544,0	7594
Cooling (AC)	Cooling (AC)	seasonal	14,29	0,0	0,0	0,0	259,4	824,1	2094,8	2822,0	2777,1	1920,4	550,4	0,0	0,0	11248
Base loads	Base loads	24h	1,50	1116,0	1015,6	1116,0	1080,0	1116,0	1080,0	1116,0	1116,0	1080,0	1116,0	1080,0	1116,0	13148
TOTAL	All loads		50,24	5718,3	5058,7	4962,8	4247,3	4618,3	5660,9	6567,9	6537,4	5849,0	4958,2	5193,4	5597,5	64969,7

Figure 3: Estimated Monthly Load Profile

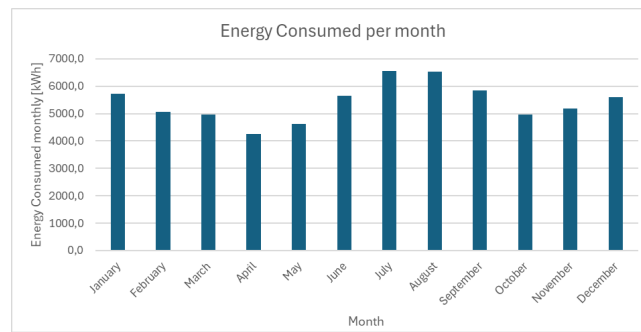


Figure 4: Energy Consumed per month

The building's energy consumption was divided into three main operational categories:

- **Workday Loads:** This includes lighting, office equipment (such as computers and printers) and the break room/kitchenette. These loads operate during normal business hours, which aligns with the period of solar generation.
- **Seasonal Loads:** The heating and cooling systems represent the major variations in the load curve. While heating increases the demand during winter, the air conditioning (AC) is used mainly during the summer. As seen in Figure 3, the building's peak consumption occurs in July (over 6500 kWh).
- **Base Loads:** This category represents the continuous energy draw of the building, regardless of the time of day or season. It includes essential infrastructure such as servers, CCTV security systems, refrigerators and other components running in standby mode. Although the instantaneous power of these devices is relatively low, their continuous operation accumulates a steady consumption of around 1,000 kWh per month. This base load is a critical factor for the project, as it represents the main energy demand that the battery system will need to sustain during the nighttime when no energy is being produced by the solar cells.

4 PV Module Selection and Geometric Sizing

4.1 PV Module Selection

For this project, a comprehensive market analysis was conducted, comparing several solar panel models, as demonstrated in Table 1 and Table 2.

Table 1: Physical specifications, price, and temperature coefficients of the different PV modules.

Model	Size and Price					Temp. Coeffs. (%/°C)		
	Length [mm]	Width [mm]	Area [m ²]	Weight [kg]	Price [€]	α_P	α_V	α_I
TW Solar 625W	2382	1134	2.70	32.5	102.78	-0.30	-0.20	+0.046
DMEGC 500W	1950	1134	2.21	26.8	102.88	-0.30	-0.30	+0.048
DAH Solar 595W	2278	1134	2.58	30.8	107.87	-0.30	-0.30	+0.046
Phono Solar 550W	2278	1134	2.58	29.0	114.65	-0.30	-0.30	+0.042
Trina Solar 430W	1762	1134	2.00	21.0	114.50	-0.30	-0.20	+0.040

Table 2: Electrical characteristics of the evaluated PV modules under STC and NOCT conditions.

Model	STC (1000 W/m ² , 25°C)						NOCT (800 W/m ² , 20°C)				
	P_{max} [W]	η [%]	I_{sc} [A]	V_{oc} [V]	I_{mp} [A]	V_{mp} [V]	P_{max} [W]	I_{sc} [A]	V_{oc} [V]	I_{mp} [A]	V_{mp} [V]
TW Solar 625W	625	23.10	16.12	49.30	15.26	40.96	472	13.01	46.93	12.32	38.32
DMEGC 500W	500	22.60	14.22	44.70	13.30	37.66	376	11.52	42.09	10.69	35.22
DAH Solar 595W	595	23.03	14.32	52.00	13.46	44.20	447	11.56	49.40	10.66	42.00
Phono Solar 550W	550	21.29	13.82	49.59	13.24	41.55	414	11.15	47.01	10.68	38.80
Trina Solar 430W	430	21.50	10.59	51.40	9.96	43.20	328	8.53	48.70	8.11	40.40

A direct economic comparison based on the price per watt (€/W) for each PV module is presented in Table 3.

Table 3: Comparison of the optimal number of panels, price per watt, and total investment per model.

Model	Pmax [W]	Unit Price [€]	Price/Watt [€/W]
TW Solar 625W	625	102.78	0.16
DMEGC 500W	500	102.88	0.21
DAH Solar 595W	595	107.87	0.18
Phono Solar 550W	550	114.65	0.21
Trina Solar 430W	430	114.50	0.27

It is possible to verify that the **TW Solar 625W** has the lowest price per watt. While the price per watt is a great economic indicator, the STC values only represent laboratory conditions. To make an accurate comparison, the performance of each model must be calculated using the hourly irradiance and temperature values, obtained from *PVGIS*.

For each solar panel model, the Power Fast Estimate was determined for each timestamp using the equation 1.

$$P_{DC}(G, T_{cell}) = P_p \cdot \frac{G}{G_{ref}} \cdot (1 - \alpha_p \cdot \Delta T) \quad (1)$$

were,

$$\Delta T = T_{cell} - 20 \quad \text{and} \quad T_{cell} = T_{air} + \frac{G}{G_{ref}}(48 - 20) \quad (2)$$

The values of V_{oc} , V_{mp} , I_{sc} and I_{mp} were also computed using the following equations:

$$V = V \cdot (1 - \alpha_v \cdot \Delta T), \quad V = V_{oc} \text{ or } V_{mp} \quad (3)$$

$$I = I \cdot \frac{G}{G_{ref}} \cdot (1 - \alpha_i \cdot \Delta T), \quad I = I_{sc} \text{ or } I_{mp} \quad (4)$$

The required number of solar panels for each hour is computed by dividing the hourly load demand (E_{load}) by the DC power output per module (P_{DC}). However, this calculation leads to an unrealistic number of panels during specific timestamps, primarily at sunrise and sunset when the load demand persists but solar irradiation is extremely low. To mitigate this issue, an appropriate statistical percentile must be applied to determine the optimal system capacity. This percentile value was computed by making an overall comparison between several key parameters, including financial indicators, total energy produced, the energy imported from and exported to the grid. This reasoning will be addressed later in this report.

The optimal percentile value was found to be 0.75, so the optimal number of solar panels was computed for each model, Table 4. By selecting the 75th percentile, the model identifies a capacity that is sufficient to cover the building's demand during 75% of the sunlit hours

Table 4: Comparison of the optimal number of panels and total investment per model

Model	Pmax [W]	Unit Price [€]	Optimal Qty.	Total Cost [€]
TW Solar 625W	625	102.78	70	7,194.60
DAH Solar 595W	595	107.87	74	7,982.38
Phono Solar 550W	550	114.65	81	9,286.65
DMEGC 500W	500	102.88	88	9,053.44
Trina Solar 430W	430	114.50	101	11,564.50

As shown in Table 4, the **TW Solar 625W** module is still the best choice. Since it has the highest power output (P_{max}) and an excellent efficiency of 23.10%, it requires the lowest number of panels (70) to reach the system's target capacity.

4.2 Geometric Sizing

After selecting the photovoltaic module to be used in the installation, the next step was to determine how many panels could be placed on the roof of Olhão's City Hall. This analysis was carried out considering the real geometry of the selected module, the chosen tilt angle, and the spacing required between rows in order to avoid mutual shading.

The selected PV module has the following dimensions:

- length: 2.382 m;
- width: 1.134 m.

In order to maximise the annual energy yield, the modules were considered with a tilt angle of 33° , which is consistent with the optimum inclination previously obtained for this location. However, tilting the panels increases the vertical height of each row and creates the possibility of shading between consecutive rows. For that reason, the row spacing must be determined geometrically.

4.2.1 Row geometry

Considering the panel length $L = 2.382$ m and the tilt angle $\beta = 33^\circ$, the vertical height of each row is given by

$$h = L \sin(\beta) \quad (5)$$

which results in

$$h = 2.382 \times \sin(33^\circ) = 1.297 \text{ m} \quad (6)$$

The horizontal projection of the panel over the roof is

$$p = L \cos(\beta) \quad (7)$$

thus,

$$p = 2.382 \times \cos(33^\circ) = 1.998 \text{ m} \quad (8)$$

To prevent shading between rows, a solar elevation angle of 29.5° was considered for the calculation. Based on this value, the shadow cast by one row over the roof is

$$s = \frac{h}{\tan(\alpha)} \quad (9)$$

where $\alpha = 29.5^\circ$. Therefore,

$$s = \frac{1.297}{\tan(29.5^\circ)} = 2.293 \text{ m} \quad (10)$$

This value represents the minimum horizontal distance required so that one tilted row does not shade the following one under the adopted solar angle.

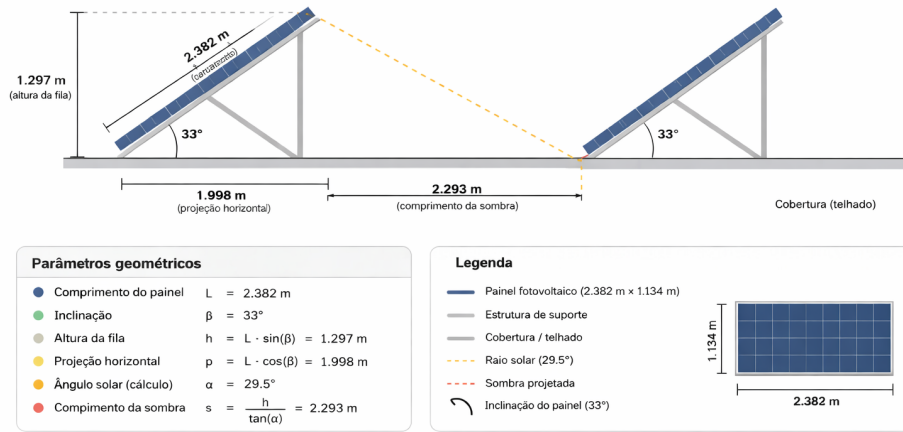


Figure 5: Geometric representation of one PV row and the spacing required to avoid shading

4.2.2 Distribution of the panels on the roof

After defining the geometric constraints, the available roof area was divided into two installation zones, denoted as Block A and Block B, according to the usable portions identified in the top view of the building. Inside each block, the rows were positioned respecting the spacing required between them.

The useful length of each row was measured directly from the available roof area, and the maximum number of modules per row was obtained by dividing that length by the module width, 1.134 m. Mathematically, the number of modules in each row is given by

$$N_i = \left\lfloor \frac{l_i}{1.134} \right\rfloor \quad (11)$$

where l_i is the available length of row i , and $\lfloor \cdot \rfloor$ represents the integer part of the result, since only complete modules can be installed.

The obtained values were the following:

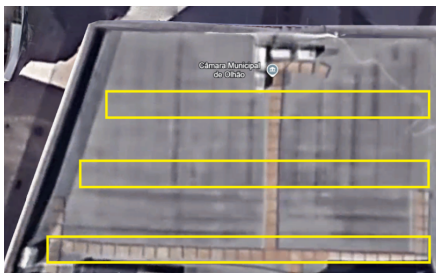
- Block A – Row 1: available length 22.7 m $\Rightarrow$ 20 modules;

- Block A – Row 2: available length 20.9 m $\Rightarrow$ 18 modules;
- Block A – Row 3: available length 19.3 m $\Rightarrow$ 17 modules;
- Block B – Row 1: available length 26.2 m $\Rightarrow$ 23 modules;
- Block B – Row 2: available length 21.7 m $\Rightarrow$ 19 modules.

Therefore, the total number of modules that can be installed on the roof is

$$N_{\text{total}} = 20 + 18 + 17 + 23 + 19 = 97 \text{ modules} \quad (12)$$

This result corresponds to the maximum number of panels that can be installed while respecting the selected inclination and the minimum spacing between rows required to limit shading effects.



(a) Block A



(b) Block B

Figure 6: Distribution of the PV rows on the available roof area

4.2.3 Maximum installation capacity of the roof

The geometric analysis carried out in this section makes it possible to determine the maximum number of photovoltaic modules that can be accommodated on the roof, considering the dimensions of the selected panel, the tilt angle of 33° , and the minimum spacing required between rows to avoid mutual shading.

Under these conditions, the roof can accommodate up to 97 modules distributed over the two available roof sections. Therefore, this value should be interpreted as the maximum feasible installation capacity from a geometric point of view, rather than as the final number of modules to be installed.

The final design may include fewer modules if that is considered more advantageous from an energetic or economic perspective. The number of installed panels can be adjusted according to the building load demand, the desired level of self-consumption, the inverter sizing, and the overall cost-effectiveness of the system.

5 Inverter Selection

5.1 PV Inverter

To convert the DC power generated by the solar modules, the **Huawei SUN2000-40KTL-M3** three-phase grid inverter was chosen. With a rated AC output of 40 kW, it is perfectly sized for the 43.75 kWp DC array, resulting in an optimal DC/AC ratio of approximately 1.09.

Table 5: DC Input parameters of the selected PV Inverter

Parameter	Value
Max. Input Voltage	1,100 V
Max. Current per MPPT	27 A (per MPPT) / 20 A (per Input)
Max. Short Circuit Current per MPPT	40 A
MPPT Operating Voltage Range	200 V ~ 1000 V
Rated Input Voltage	600 V
Number of Inputs	8
Number of MPP Trackers	4

The inverter features 4 independent Maximum Power Point Trackers (MPPTs), which allows for an optimised distribution of the 70 PV modules. To respect the inverter's voltage and current constraints, the array is divided into four strings, one per MPPT:

- MPPT 1: 1 array of 18 solar panels in series
- MPPT 2: 1 array of 18 solar panels in series
- MPPT 3: 1 array of 17 solar panels in series
- MPPT 4: 1 array of 17 solar panels in series

Under STC, the nominal operating voltage (V_{mp}) for the longest string (18 modules) is calculated as:

$$V_{mp_array} = 18 \times 40.96 \text{ V} = 737.28 \text{ V}$$

This value aligns with the inverter's optimal MPPT operating voltage range and is close enough to the rated input voltage (600V). Since there are no parallel connections, the produced current in each array is equal to the current produced by one individual cell, 15.26A, which is within the MPPT maximum input current.

5.2 Battery Pack Inverter

Since this building operates in a three-phase electrical configuration, it is necessary to have one battery inverter for each phase. Therefore, a pack of three **Victron MultiPlus-II 48/3000** bidirectional inverters was selected. During periods of excess solar generation, these inverters convert AC power to DC to charge the 48V battery pack. Conversely, during nighttime, they invert the DC battery power back to AC to meet the energy demand.

6 Battery Selection

The sizing of the Battery System is dictated by the base loads during the night. To ensure complete energy autonomy overnight, the system must provide 1.8kW for a duration of 15 hours. This results in a total energy of 27kWh.

A comparative analysis of different 51.2V LiFePO4 batteries was performed, as demonstrated in Table 6.

Table 6: Comparison of selected batteries

Battery Model	Nominal Voltage (V)	Capacity (kWh)	Discharge Current (A)	Price (€)
Eco-Worthy PowerMega 314	51.2	16	200	1849
Deye SE-F12-C	51.2	11.8	230	1600
WECO 16k0	51.2	16.06	320	3299
FLB48314TG1	51.2	16	160	2310

The **Eco-Worthy PowerMega 314** was selected as the optimal solution because, for the price, it provides great capacity relative to the other models.

To satisfy the 27 kWh requirement, it is necessary to have two units in parallel, providing a total storage capacity of 32 kWh. Since the battery pack has a capacity of 32kWh and the estimated overnight power is 27kWh, their percentage before the sunrise is approximately 15%, so it is possible to avoid deep discharge cycles to prevent the battery pack degradation over time.

According to the datasheet, the battery is rated for a lifespan of up to 10,000 cycles. Given that the estimated annual usage is approximately 248 cycles, this battery pack will operate throughout the entire duration of the project.

7 Cable Sizing

To select the appropriate cabling for the PV system, it is essential to consider the maximum current generated, the thermal ampacity of the cables, and the allowable voltage drop across the circuit. The cable ampacity ensures that the selected cross-section can safely carry the maximum expected current without degrading the cable's insulation, especially under severe environmental conditions like summer temperatures. Limiting the voltage drop is also important to minimise Joule losses and maximise the total energy transmitted from the solar panels to the inverter.

For this project, a 4mm² cross-section cable was selected. According to the manufacturer's datasheet, the cable has a maximum current capacity of 55 A under laboratory conditions and an operating temperature range of -40°C to +90°C. Although the nominal ampacity of the cable is 55 A, this capacity decreases as the temperature rises due to direct sun exposure on the rooftop. Given that the selected TW Solar 625W modules have a maximum power current (I_{mp}) of 15.26 A and a short-circuit current (I_{sc}) of 16.12 A, the 4 mm² cross-section operates with a safety margin, completely eliminating the risk of overheating.

The voltage drop must be evaluated to ensure it remains below the industry standard of 1%. The worst-case scenario occurs at the furthest module, located approximately 40 meters from the inverter, resulting in an 80-meter round-trip circuit. The voltage drop can be computed using the equation

$$\Delta V = \frac{2 \cdot L \cdot I \cdot R}{1000} \quad (13)$$

where L is the cable length, I is the operating current, and R is the cable resistance for each kilometre, which is 4.75 Ω/km. This results in a voltage drop of 6.1256V, which represents a voltage drop of 0.83%.

To connect all the solar panels in four different arrays, it is necessary to use 195 meters of cable. Since the price per 25 meters is 94€, the total price will be approximately 752€.

8 Full System Diagram

With all the components selected, the full circuit diagram was created.

The system is designed as a hybrid AC-coupled architecture, for maximum reliability managing solar production.

The production is based on 70 625W PV modules, outputting DC voltage. In order to connect the PV modules to the Electrical Panel the inverter Huawei 40KTL is used, converting DC into AC. The smart meter Huawei is critical, as it monitors the energy flow and, if required, communicates with the inverter to limit the export of energy, preventing grid export and prioritising local consumption and battery charging.

For the energy storage and management, Eco-Worthy batteries are installed and connected to the electrical panel through 3 Victron Multiplus-II inverters, one for each phase of the electrical panel. These inverters allow bidirectional flow, charging the batteries when production is higher than consumption and discharging them when needed. These operations are controlled by a Cerbo GX, a device that acts as the brain of the system, gathering data from the Victron Smart Meter, which is responsible for seeing if the batteries need charging or discharging.

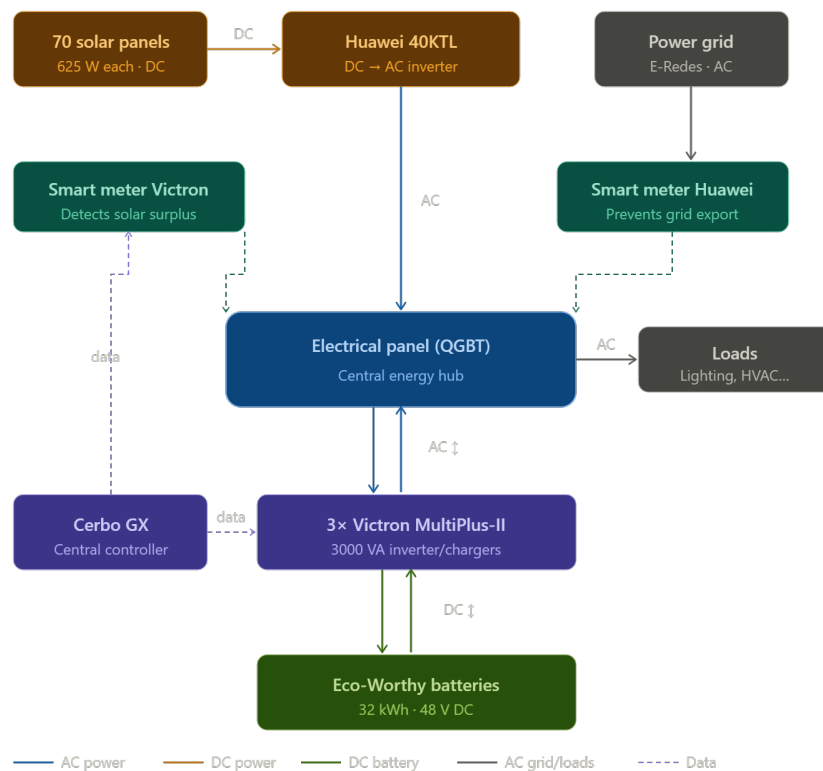


Figure 7: Full System Diagram

9 Excel Calculation Methodology

All the calculations for this project were performed using Excel. In this section, the process of all the computations is going to be described.

PVGIS was used to get hourly values of irradiance at the optimised angle and temperature at the site location. With this, an Excel table was created containing timestamps, irradiance, temperature

and energy demand for each hour from the first day of 2005 to the last day of 2023. After this, the following parameters were computed for each timestamp:

- Temperature of the cell:

$$T_{cell} = T_{air} + \frac{G}{800} \times (NOCT - 20)$$

- Power output of one PV module:

$$P_{module} = P_{ref} \times \left(\frac{G}{800} \right) \times [1 + \gamma \times (T_{cell} - 20)]$$

- V_{oc} :

$$V_{oc} = V_{oc,ref} \times [1 - \beta \times (T_{cell} - 20)]$$

- V_{mp} :

$$V_{mp} = V_{mp,ref} \times [1 - \beta \times (T_{cell} - 20)]$$

- I_{sc}

$$I_{sc} = I_{sc,ref} \times \left(\frac{G}{800} \right) \times [1 + \alpha \times (T_{cell} - 20)]$$

- I_{mp}

$$I_{mp} = I_{mp,ref} \times \left(\frac{G}{800} \right) \times [1 + \alpha \times (T_{cell} - 20)]$$

- Number of solar panels for specific hour:

$$N_{panels,h} = \frac{P_{load,h}}{P_{module,h}/1000}$$

This gives the number of solar panels needed to supply the demand of a specific hour given the irradiance at that hour.

- Number of PV modules chosen:

$$N_{optimal} = \lceil \text{Percentile}(N_h : N_h > 0, P) \rceil$$

This formula uses a statistical percentile to determine a system size that covers the building's load for a specific percentage of the sunlit hours. By filtering out nighttime (zero values) and rounding up, it identifies the exact number of panels needed to reach the 75% target without under-sizing.

- Maximum power output from the inverter:

$$P_{AC,h} = \frac{N_{panels} \times P_{module,h} \times \eta_{inv}}{1000}$$

This gives the total AC power available to the building at a specific hour in kW. N_{panels} is the number of panels computed in the previous formula, $P_{module,h}$ is the power produced by a single module in a specific module, η_{inv} is the inverter efficiency, considered to be 98%.

- Battery Power Flow: This formula determines the hourly energy flow to or from the battery system. It operates based on two scenarios:

- **Charging:** If solar production ($P_{AC,h}$) exceeds the load, the battery charges using the surplus, limited by the remaining available capacity ($C_{max} - E_{stored,h-1}$).
- **Discharging:** If the load exceeds solar production, the battery discharges to cover the deficit, limited by the energy currently stored and the minimum safety threshold (SOC_{min} , considered as 10% of total capacity).
- **Energy stored on Battery:** The battery is considered to be installed at a 30% capacity, corresponding to 9,6 kWh. From this hour onwards, battery capacity is computed using the formula:

$$E_{stored,h} = E_{stored,h-1} + E_{bat,h}$$

- **State of Charge (SOC %):**

$$SOC_h = \frac{E_{stored,h}}{C_{nominal}} \times 100$$

- **Grid Interaction ($P_{grid,h}$):**

$$P_{grid,h} = P_{load,h} - P_{AC,h} + E_{bat,h}$$

If the value is positive, the building is buying energy from the grid. If negative, the building is producing more than what it needs.

- **Battery Cycles:**

$$Cycles_h = \begin{cases} \frac{|E_{bat,h}|}{C_{nominal}} & \text{if } E_{bat,h} < 0 \\ 0 & \text{if } E_{bat,h} \geq 0 \end{cases}$$

These calculations were performed for every line of the Excel and were important to make the next sections. Below is presented a part of the table to better illustrate what was performed on the original Excel. This table corresponds to the first day of the calculations, January 1st 2005.

Table 7: Hourly Energy Balance Results (Example: January 1st 2005)

Time (hh:mm)	Irradiance (W/m^2)	P_{load} (kW)	$P_{AC,inv}$ (kW)	E_{bat} (kW)	Stored (kWh)	SOC (%)	Grid (kW)	Cycles (count)
00:00	0.00	1.66	0.00	-1.66	9.60	30%	0.00	0.05
04:00	0.00	1.66	0.00	-1.43	3.20	10%	0.23	0.25
05:00	0.00	1.66	0.00	0.00	3.20	10%	1.66	0.25
09:00	325.58	5.57	13.10	7.51	10.71	33%	0.00	0.25
11:00	805.02	6.73	30.50	4.43	32.00	100%	-19.32	0.25
12:00	919.37	7.66	34.30	0.00	32.00	100%	-26.66	0.25
17:00	22.42	1.91	0.90	-0.99	31.01	97%	0.00	0.28
20:00	0.00	1.69	0.00	-1.69	25.94	81%	0.00	0.44
23:00	0.00	1.69	0.00	-1.69	20.87	65%	0.00	0.60

It is important to note that at 9AM the solar irradiance is higher than 0 and the PV modules are producing 13.1 kW but the building is only demanding 5.57 kW. As expected the grid import is zero, meaning the building is being fully supplied by the solar system and the surplus of energy (13.1 kW - 5.57 kW = 7.51) is going to the battery and charging it.

At 8PM, the irradiance is zero, so there is no solar production, and the building is demanding 1.69 kW. Because the batteries are charged all day, they are capable of supplying the building's demand.

10 Monthly Energy Balance and Performance Analysis

To better analyse the hourly data, a monthly energy production and consumption analysis table was obtained, Table 8. This table aggregates the hourly values into monthly totals, allowing for a clear evaluation of the system's performance across different seasons. By observing these results, it is possible to identify periods of energy surplus and the effectiveness of the battery storage system in reducing grid dependency. Key indicators such as the number of battery cycles and self-sufficiency ratios are presented to validate the technical viability of the proposed 70-panel configuration.

Self sufficiency and self consumption ratios were also determined, contributing to better analyse the percentage of energy consumed that was actually produced by the solar panels and the percentage of energy produced that was consumed by the building, respectively.

Table 8: Detailed Monthly Energy Balance including Storage System

Parameter [kWh]	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	Total
Pmax Inv (Prod.)	5424	5588	6908	7351	8068	7955	8521	8296	7413	6351	5245	4952	82072
Eload (Demand)	5718	5059	4963	4247	4618	5661	6568	6537	5849	4958	5193	5598	64970
Grid Import	1296	751	458	166	98	121	95	133	268	550	1011	1541	6486
Grid Export	996	1279	2398	3268	3546	2415	2048	1895	1836	1946	1065	895	23587
Batt. Discharge	566	621	700	682	640	648	746	804	744	687	620	480	7938
Monthly Cycles	17.7	19.4	21.9	21.3	20.0	20.2	23.3	25.1	23.2	21.5	19.4	15.0	248.1
Self Sufficiency	77%	85%	91%	96%	98%	98%	99%	98%	95%	89%	81%	72%	90%
Self Consumption	82%	77%	65%	56%	56%	70%	76%	77%	75%	69%	80%	82%	72%

With this data, it is possible to perform various figures to analyse the system. Figure 8 shows the difference between the energy produced by the solar panels and the energy demanded by the building.

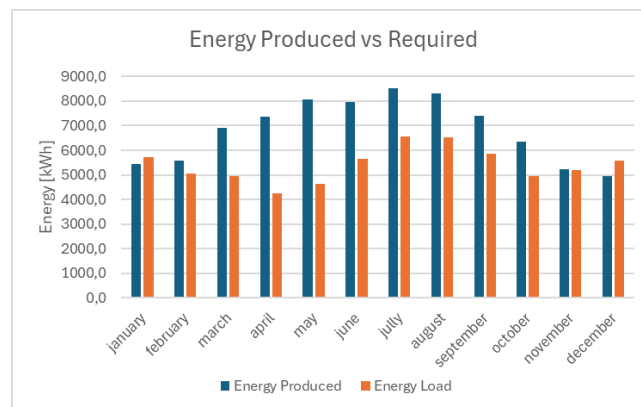


Figure 8: Energy Produced vs Required

It is possible to note that production is always higher than demand, with January and December being an exception, where solar irradiance is not high enough to cover the total demand. This results in a production of energy slightly lower than the demand. On the other hand, months with high irradiance, like the summer months, exhibit a significant energy surplus.

The next graph demonstrates the effectiveness of the system in achieving self-sufficiency. The energy consumed every month is divided among the 3 sources that can power the building. The grid in blue, the batteries in orange and the solar panels in green.

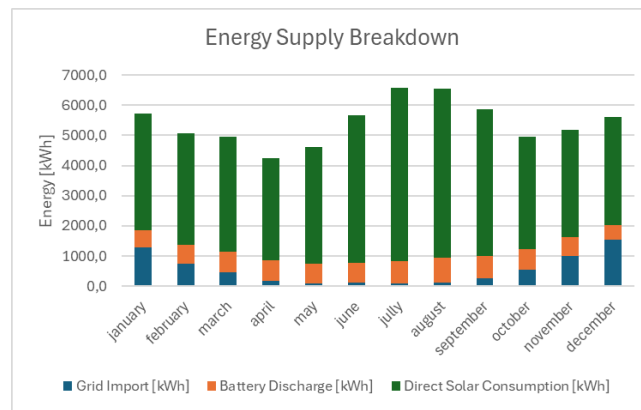


Figure 9: Energy Supply Breakdown

It is possible to see that the vast majority of the energy demanded by the building is delivered by the solar panels, with only a small fraction being imported from the grid. In the summer months, the energy supplied by the grid decreases due to the high energy that the PV system can deliver because of the high irradiance. It is important to note that the energy supplied by the batteries is almost constant every month. This is because the batteries always supply the night load that is constant throughout the year.

The next graph demonstrates the evolution of performance indicators throughout the year.

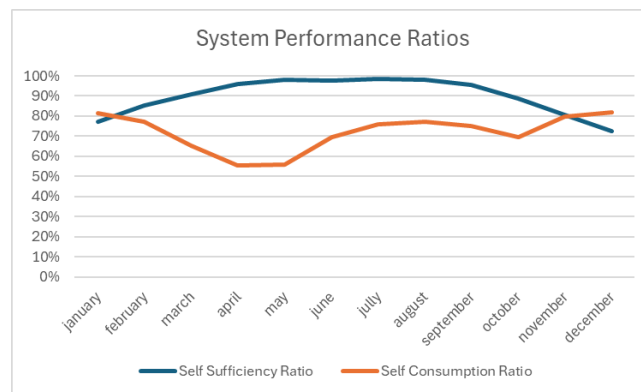


Figure 10: System Performance Ratios

Self sufficiency ratio is always higher than 70% with values reaching almost 100% on the summer months, once more due to the high energy the solar panels can produce. Self consumption ratio reaches its lowest in the spring months, partially due to the lack of thermal loads, as neither heating nor intensive air conditioning is required during this period. This results in a higher energy being wasted. Since a smart meter is installed in the circuit, it detects when production is greater than consumption. It then commands the inverter's MPPT to operate at an adjusted, lower-power point, perfectly matching the energy production to the building's power demand.

11 Financial Analysis

Photovoltaic systems are a big investment, but are designed to pay themselves off through the consistent reduction of monthly electricity expenses in electricity. In order to analyse whether this project is economically viable, a comprehensive analysis was carried out.

In this section, the analysis is detailed, and the financial indicators are computed.

The first step is to determine the total investment in order to have the system fully functional. This includes the solar panels, batteries, inverters, etc... The number of solar panels chosen will be explained in the next section. Table 9 demonstrates the components chosen and the respective prices.

Table 9: Initial Investment Cost Breakdown

Component	Quantity	Unit Price (€)	Total Price (€)
Solar Panels	70	102.78	7194.60
Eco-Worthy PowerMega 314 (51.2V, 314Ah, 16kWh)	2	1849.00	3698.00
Huawei SUN2000-40KTL-M3 Inverter	1	2987.00	2987.00
Victron MultiPlus-II 48/3000	3	550.00	1650.00
Huawei Smart Power Sensor DTSU666-H	1	152.87	152.87
Victron Cerbo GX	1	223.00	223.00
Victron Energy Meter ET340	1	79.00	79.00
Structures	1	3800.00	3800.00
Cabling	1	752.00	752.00
Installation	1	3000.00	3000.00
Initial Investment			23536.47

Secondly, the analysis requires the definition of several constant parameters. The project lifetime is considered to be 25 years, which is the standard for PV systems. Annual Maintenance is considered to be 800€. These values are presented in the Table 10.

Table 10: Economic and Technical Parameters

Parameter	Value	Unit
Grid Purchase Price	0.10	€/kWh
Discount Rate (DR)	4.00	%
Project Lifetime (n)	25	Years
Annual Maintenance (O&M)	800.00	€/Year
Solar Panel Degradation Rate	0.50	%

The financial viability of the system is detailed in Table 11, which presents a 25-year cash flow projection. The initial investment (year 0) is the total investment obtained in Table 9.

From year 1 onwards, a fixed annual O&M cost is accounted to ensure system performance. Savings each year were computed by multiplying the grid purchase price by the amount of kWh saved. The PV modules are considered to have a 0.5% annual degradation, leading to a slight reduction of savings over time. The net flow is the actual liquidity generated by the project each year. It is computed by subtracting the annual O&M costs from the savings. Accumulated flow tracks the total financial performance of the investment over time, summing every net flow from year 0. With this column, it is possible to see the year when the investment pays for itself.

Table 11: Project Cash Flow Analysis (25-year lifetime)

Year	Investment (€)	O&M (€)	Savings (€)	Net Flow (R_t) (€)	Acc. Flow (€)
0	23,536.47	0.00	0.00	-23,536.47	-23,536.47
1	0.00	800.00	5,848.46	5,048.46	-18,488.01
2	0.00	800.00	5,819.21	5,019.21	-13,468.80
3	0.00	800.00	5,790.12	4,990.12	-8,478.68
4	0.00	800.00	5,761.17	4,961.17	-3,517.52
5	0.00	800.00	5,732.36	4,932.36	1,414.84
⋮	⋮	⋮	⋮	⋮	⋮
12	0.00	800.00	5,534.71	4,734.71	35,146.81
⋮	⋮	⋮	⋮	⋮	⋮
25	0.00	800.00	5,185.55	4,385.55	94,229.46

With this data, the following financial indicators were computed.

11.1 Financial Indicators

Net Present Value - NPV

This indicator represents the difference between the present value of cash inflows and the present value of cash outflows, over a period of time, determining an investment's profitability. If $NPV < 0$, the project is expected to result in a net loss for the investors. If $NPV > 0$, the investors should expect profit. If $NPV = 0$, the project is not expected to bring profit or loss, so the investors should make a decision about the project only considering non-monetary factors.

NPV can be obtained with the following formula:

$$NPV = \sum_{t=0}^n \frac{R_t}{(1 + DR)^t} - I_0$$

Where R_t is the revenue, DR is the discount rate applied to the annual cash flows until the end of the project's lifetime, n and I_0 is the initial investment.

Given our values

Payback Period - PP

This indicator gives the investor the time it takes to recover the cost of the investment. It can be obtained with the following formula:

$$PP = A + \frac{B}{C}$$

Where A is the last year with a negative cumulative cash flow, B is the absolute value of cumulative cash inflows at the end of year A and C is the cash flow generated in the year immediately after year A.

Levelized Cost of Energy - LCOE

This indicator describes the cost of each unit of power generated over the lifetime of the project. This indicator does not account for interest rates and does not include the discount to present value. It can be computed using this formula:

$$LCOE = \frac{I_0 + \sum_{t=0}^n O\&M_t}{\sum_{t=0}^n E_t}$$

Where I_0 is the initial investment, $O\&M_t$ is the operations and maintenance costs during a single period t , n is the number of time periods that correspond to the project's lifetime, and E_t is the total energy generated during a single period t .

Return on Investment - ROI

This indicator is a performance measure used to evaluate the profitability of the investment over the project's lifetime. It is computed as:

$$ROI = \frac{\text{Net benefit}}{\text{Total investment}}$$

11.2 Cost-Optimal System Sizing

The number of solar panels depends on the financial indicators. As explained before on the report, the number of solar panels is determined by selecting the optimal design percentile — a metric that scales the system to cover a specific proportion of the building's energy demand. If a low percentile is chosen, the initial investment is lower but the system will be undersized and unable to generate the energy demand of the building. On the other hand, if the percentile is higher, the initial investment is high and the system might be oversized, leading to an excess of energy being produced that can't be used by the building, not contributing to good financial indicators.

To select a good percentile, a *What-if Analysis* was performed on Excel. For each percentile, the table computed several factors that help to choose the best percentile. The results obtained are presented below, as well as the evolution of the financial indicators for different percentiles.

Table 12: System Sizing Optimisation and Sensitivity Analysis

Percentile	NPV (€)	Payback (yrs)	LCOE	PV Panels	Total Prod. (kWh)	Grid Import (kWh)	Grid Export (kWh)	Self-Suff. Ratio	Self-Cons. Ratio
0.30	9,508.55	9.98	0.06	24	28,138.95	37,518.87	688.11	43%	98%
0.35	15,569.67	8.29	0.05	29	34,001.23	33,096.64	2,128.11	50%	94%
0.40	27,024.68	6.41	0.04	39	45,725.79	24,700.93	5,456.33	63%	88%
0.45	33,630.12	5.73	0.03	45	52,760.53	19,843.47	7,633.55	70%	86%
0.50	38,430.04	5.35	0.03	50	58,622.81	16,269.50	9,921.81	75%	83%
0.55	42,773.50	5.06	0.03	55	64,485.09	13,002.53	12,517.07	80%	81%
0.60	45,948.25	4.89	0.03	59	69,174.91	10,590.74	14,795.06	84%	79%
0.65	47,942.78	4.79	0.02	62	72,692.28	9,041.88	16,763.56	86%	78%
0.70	49,531.96	4.73	0.02	65	76,209.65	7,765.64	19,004.69	88%	76%
0.75	50,921.27	4.71	0.02	70	82,071.93	6,485.58	23,586.91	90%	72%
0.80	51,996.75	4.79	0.02	80	93,796.49	5,070.95	33,896.84	92%	65%
0.85	52,254.48	4.95	0.02	93	109,038.42	3,998.95	48,066.77	94%	57%
0.90	50,905.08	5.50	0.01	129	151,246.84	2,417.92	88,694.16	96%	42%

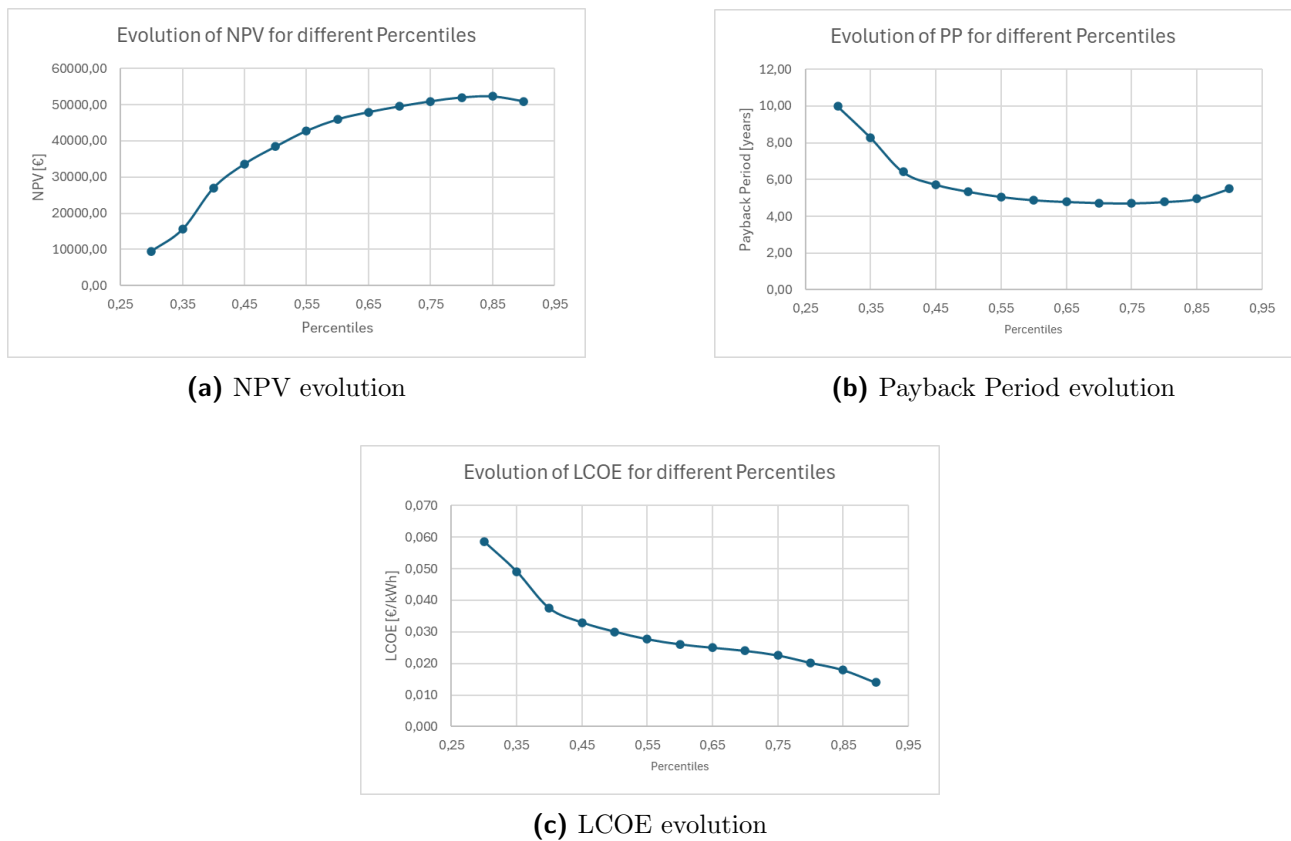


Figure 11: Economic performance indicators for different percentiles

It is possible to see that NPV has an upward trend with the increasing of solar panels. However, it reaches a peak around 0.85 and starts descending from there. From this point onwards, the initial investment is very high, and it starts to be not relevant to install so much production that cannot be used by the building. Payback Period decreases with the introduction of more solar panels, but it reaches a minimum at the percentile 75%, and LCOE is always decreasing. This happens because the high fixed costs of the system are divided by a much larger amount of generated energy. Even though the energy becomes cheaper to produce on paper, a large portion of it is exported to the grid without generating a financial return.

Taking this into consideration, the percentile chosen was 75% as this offers the best balance between NPV, PP and LCOE. Choosing this percentile means the installation of 70 PV modules. The final financial indicators are presented in the table below.

Table 13: Financial Indicators

Financial Indicators	Value
NPV (€)	50,921.27
Payback Period	4.71
LCOE	0.023
ROI	500%

These values show that the investment is viable and can reduce the electricity costs. The payback period is inside the 5-year mark, that represent a highly attractive return on investment for a project of this scale.

12 Conclusion

This project aimed to install a PV system on the Olhão's City Hall with the goal of reducing the electricity bill. The project started with extracting data from PVGIS of irradiance and ambient temperature for every hour. With this data, it was possible to compute important parameters of the system for every hour, like the total energy being produced and the remaining energy that would be necessary for the building.

Different solar panels were compared to see which one would be the best. The module chosen was the TW Solar 625W, as it was the one with the lowest price per watt produced. All the computations to see the best configuration of modules on the roof were also performed, with 97 modules being the maximum to be installed. For the inverters, the Huawei SUN2000-40KTL-M3 was selected to ensure the connection of the solar modules to the Electrical Panel. For the battery's side, the Victron Multiplus-II was selected to ensure charging and discharging of the batteries to supply the building demand. The batteries selected were the Eco-Worthy PowerMega 314, with a total capacity of 32 kWh, being able to supply the building for one entire night. Cables were also sized according to the requirements of the system and voltage drop. The final price came around 752€.

With all the components selected, a full diagram of the circuit was also presented in order to better visualise the system's architecture and the electrical integration between the PV array, the hybrid inverter, and the battery storage system.

A monthly energy balance and performance analysis was also performed, with data from all 19 years divided into each month. It was possible to see that the energy produced is almost always higher than the energy load, with the summer months being the ones with more production. Over one entire year, the system is 90% self sufficient, meaning that only 10% of the energy being consumed comes from the grid.

Finally, a financial analysis was performed. A project cash flow analysis was computed and it was possible to see that with an initial investment of 23536.47€, the NPV was 50921.27€, meaning that in 25 years the project would generate around 50k € of revenue in today's money. The payback period was around 4 years and 8 months and the project reduced the cost of energy to around 0.02€/kWh. Comparing with the price of 0.10€/kWh that the city hall would have to pay, this is a 80% decrease in the average cost of energy (LCOE) for the building.

To conclude, the project was well executed and successfully achieved its primary goal: to design a sustainable energy system that balances technical performance with economic profitability, resulting in an 80% reduction in energy costs and a fast return on investment.